Multimodal Emotion Recognition from EEG and Facial Expressions using a Two-Stage Sugeno Fuzzy Inference System

Soft Computing Project Report

Dataset: DEAP • Pipeline: MLDC → Late Fusion → Sugeno FIS

Idea in one line: train the best EEG classifier and a fine-tuned face CNN separately, turn their outputs into crisp probabilities, and let a Sugeno fuzzy system decide *how much to trust each modality* before producing the final emotion.

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1 Overall Architecture

- **Two independent branches** — EEG and face are processed and classified separately (late fusion).
- **EEG branch** — handcrafted features + classical ML — strong on small, tabular EEG feature sets.
- **Face branch** — a pretrained CNN fine-tuned end-to-end — learns expression features directly from pixels.
- **Crisp interface** — each branch outputs a calibrated probability vector over 4 emotions.
- **Two-stage Sugeno fusion** — FIS-1 estimates trust in EEG; FIS-2 scores every emotion using that trust.
- **Decision** — the emotion with the highest fuzzy score wins, or "Uncertain" if the score is too low.
- **MLDC flowchart (Fig. 1)** — the step-by-step development cycle.
- **Architecture (Fig. 2)** — the deployed system from raw data to final emotion.

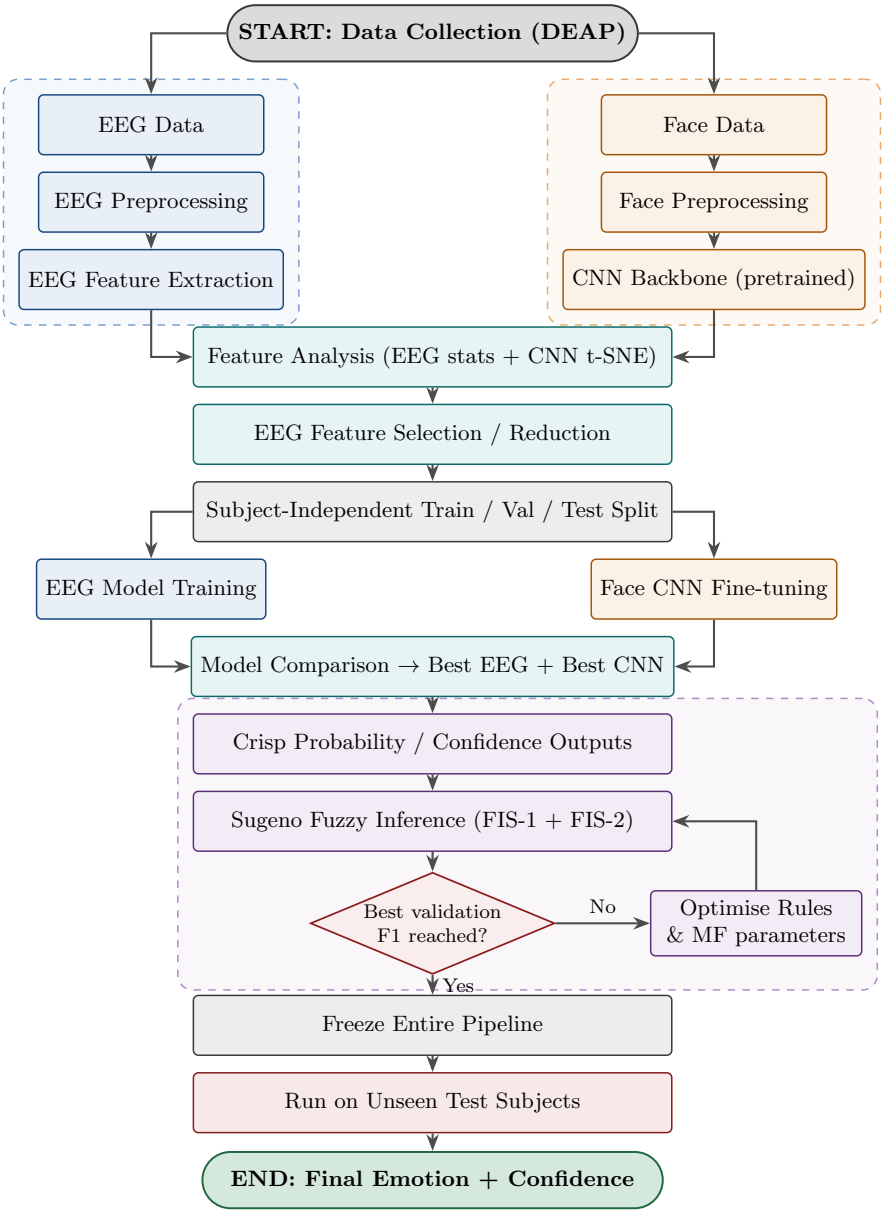


Figure 1: MLDC flowchart. Parallel branches merge for analysis and again for fusion; the Sugeno system is tuned in a validation loop before the pipeline is frozen.

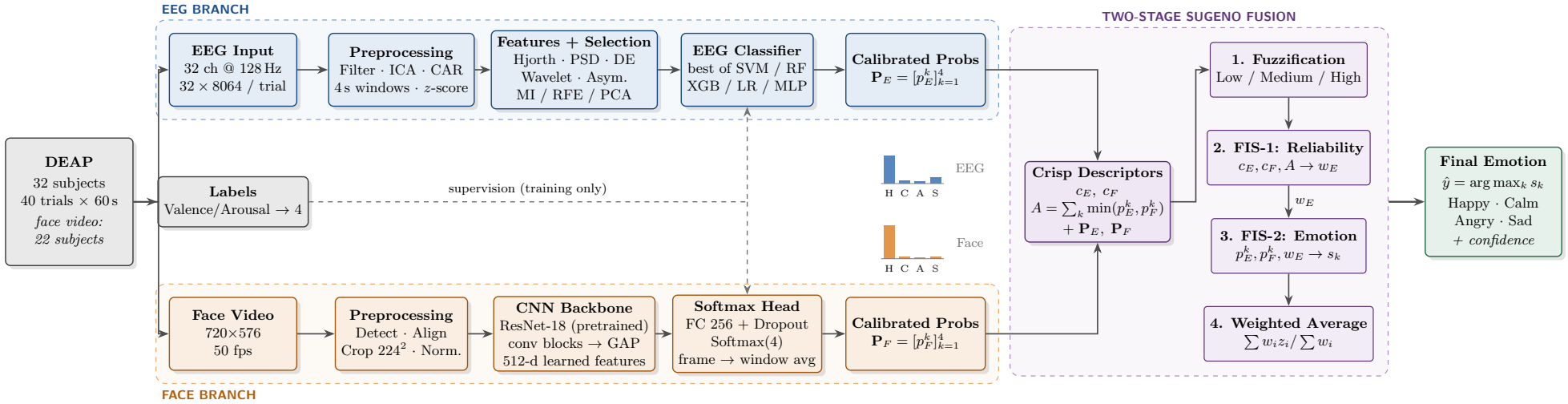
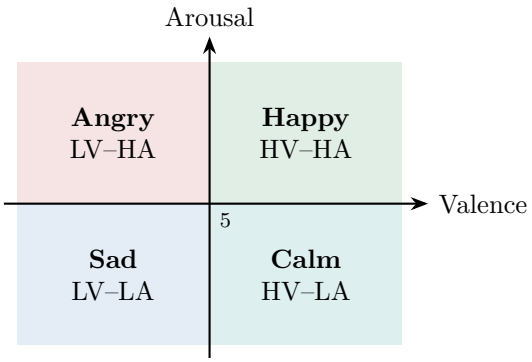


Figure 2: End-to-end architecture. Two independent branches produce crisp probability vectors; a two-stage Sugeno system converts them into trust weights and per-emotion scores.

2 Data Gathering — DEAP Dataset

- **Subjects & trials** — 32 participants each watched 40 one-minute music-video clips.
- **EEG** — 32 channels, 512 Hz raw (128 Hz in the preprocessed release).
- **Face video** — frontal face recorded for 22 of the 32 participants — only these are usable for fusion.
- **Labels** — self-rated Valence, Arousal, Dominance, Liking on a 1–9 scale per trial.
- **Pairing** — EEG and face come from the same subject and same trial, so the modalities align naturally.
- **Access** — free for academic use after signing the DEAP EULA.

Label mapping. DEAP has no discrete emotion labels, so the valence–arousal plane (threshold = 5) is split into four classes:



- **Quadrant labels** — converts continuous ratings into 4 classes usable by classifiers.
- **Class imbalance** — quadrants are not equally populated, so macro-F1 is reported, not just accuracy.

3 Data Preparation

3.1 EEG Signal Preprocessing

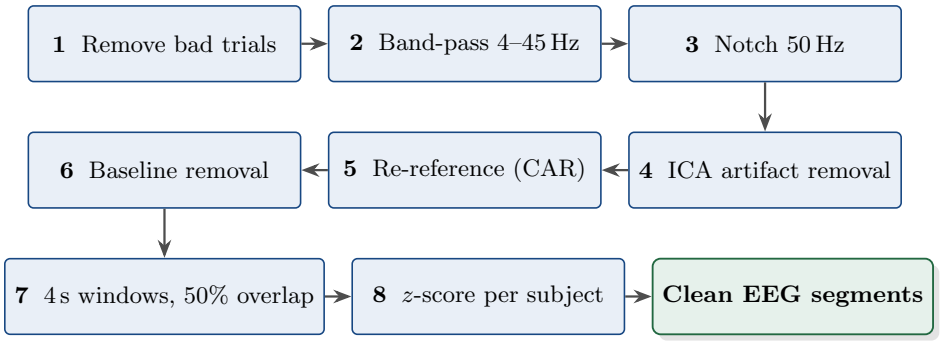


Figure 3: EEG preprocessing pipeline.

- **Remove bad trials** — drops recordings with flat, saturated or disconnected channels.
- **Band-pass filtering (4–45 Hz)** — removes noise outside the frequency range of interest.
- **Notch filtering (50 Hz)** — removes power-line interference.
- **ICA artifact removal** — separates and deletes eye-blink, muscle and ECG components.
- **Re-referencing (CAR)** — subtracts the average of all channels to remove common noise.

- **Baseline removal** — discards the 3 s pre-trial baseline so only stimulus response remains.
- **Segmentation** — splits each 60 s trial into 4 s windows, giving more training samples.
- **Normalisation** — z-scores each subject so amplitude differences between people do not dominate.
- **Label assignment** — every window inherits its trial's quadrant label.

Note: steps 2, 3, 4 and downsampling are already applied in DEAP's preprocessed release.

3.2 Facial Image Preprocessing

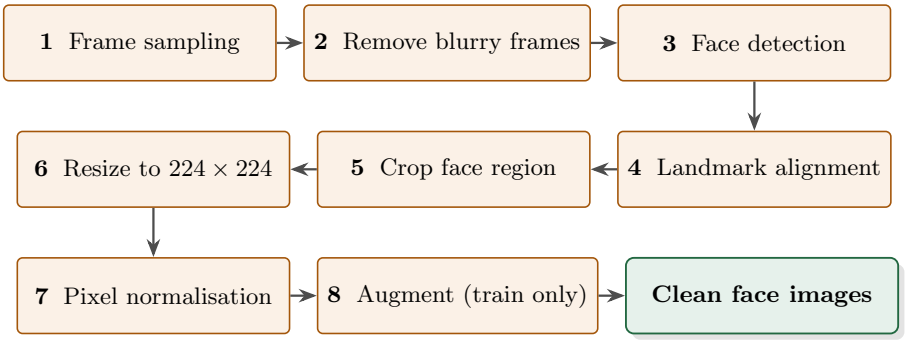


Figure 4: Face preprocessing pipeline.

- **Frame sampling** — picks frames inside each 4 s EEG window so both modalities stay time-synchronised.
- **Remove poor frames** — drops blurred, occluded or face-missing frames.
- **Face detection (MTCNN)** — locates the face bounding box in each frame.
- **Landmark alignment** — rotates the face so the eyes are horizontal.
- **Crop** — keeps only the face, removing background.
- **Resize** — gives every image the same input size required by the CNN.
- **Normalisation** — scales pixels to the range expected by the pretrained network.
- **Augmentation** — small rotation, flip and brightness changes — vital because the CNN sees only 15 training subjects.
- **ImageNet normalisation** — uses the mean/std the pretrained CNN was trained with.
- **Label assignment** — each frame inherits the label of its trial.

4 Feature Extraction

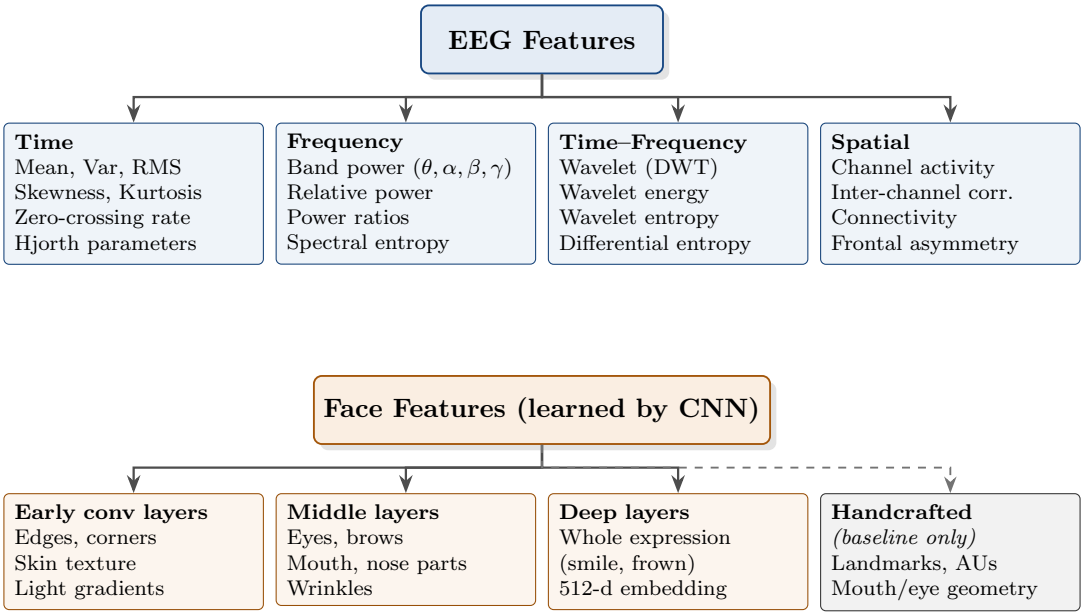


Figure 5: Feature taxonomy for both modalities.

4.1 EEG Features

- **Mean / Variance / RMS / Std** — capture signal level and energy of each channel.
- **Skewness / Kurtosis** — describe asymmetry and peakedness of the amplitude distribution.
- **Zero-crossing rate** — rough estimate of the dominant frequency.
- **Hjorth parameters** — activity, mobility and complexity of the signal in three numbers.
- **Band power (PSD via Welch)** — energy in each brain rhythm — the strongest emotion cue.
- **Relative power & ratios** — e.g. β/α — normalised and robust to amplitude scale.
- **Spectral entropy** — measures how spread out (irregular) the spectrum is.
- **Wavelet energy / entropy** — captures transient changes that FFT misses.
- **Differential entropy (DE)** — log-variance per band; a top-performing EEG emotion feature.
- **Frontal asymmetry** — left–right alpha difference (e.g. F4–F3); linked to valence.
- **Connectivity** — correlation between channels shows how brain regions co-activate.

Band	Range (Hz)	Usable in DEAP	Associated state
Delta δ	1–4	raw data only	deep sleep
Theta θ	4–8	✓	drowsiness, emotion processing
Alpha α	8–13	✓	relaxed, calm
Beta β	13–30	✓	alert, active thinking
Gamma γ	30–45	✓	high arousal, cognitive load

- **Delta limitation** — the 4 Hz high-pass in preprocessed DEAP removes delta, so it is only available from raw data.

4.2 Face Features — Learned by a CNN

- **Why CNN, not SVM/XGBoost** — images are raw pixels; a CNN learns its own expression features instead of relying on hand-picked ones.

- **Hierarchical features** — early layers learn edges, deeper layers learn eyes/mouth shapes and full expressions.
- **Transfer learning** — start from a CNN pretrained on ImageNet / VGGFace2 — 15 subjects are too few to train from scratch.
- **Freeze early layers** — generic edge/texture filters are kept; only deeper layers adapt to emotions.
- **Global Average Pooling (GAP)** — compresses the final feature map into a 512-d vector with no extra parameters.
- **No manual feature selection** — the CNN learns a compact representation itself, so MI/RFE/PCA apply only to EEG.
- **Handcrafted baseline** — landmarks + AUs with SVM are kept only to prove the CNN is better.

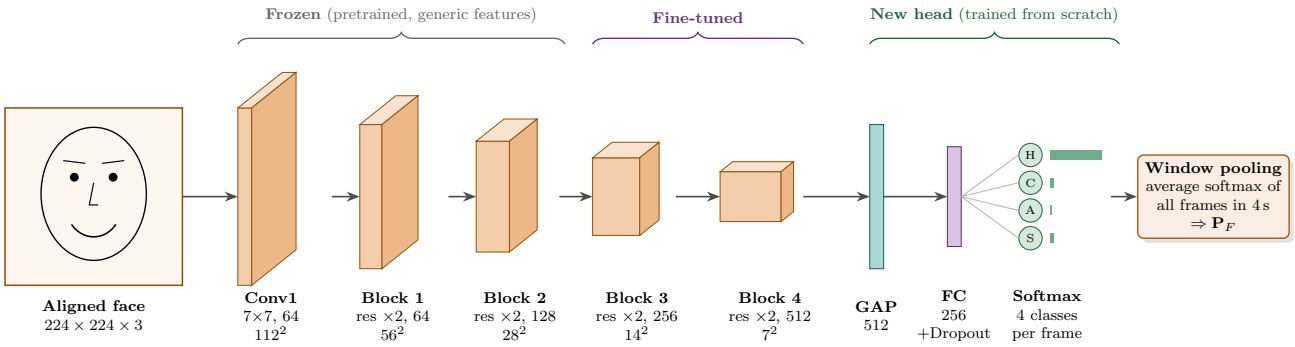


Figure 6: Face CNN (ResNet-18 transfer learning). Early blocks stay frozen, deep blocks are fine-tuned, and a new softmax head outputs 4 emotion probabilities per frame, averaged over each 4 s window.

- **Conv layers** — slide learnable filters over the image to detect local patterns.
- **Residual blocks** — skip connections let gradients flow, so deeper networks train stably.
- **Dropout** — randomly switches off neurons during training to reduce overfitting.
- **Softmax** — turns the 4 output scores into probabilities that sum to 1.
- **Window pooling** — averaging frame probabilities gives one $\mathbf{P}_F$ per EEG window, keeping the modalities aligned.

5 Feature Analysis

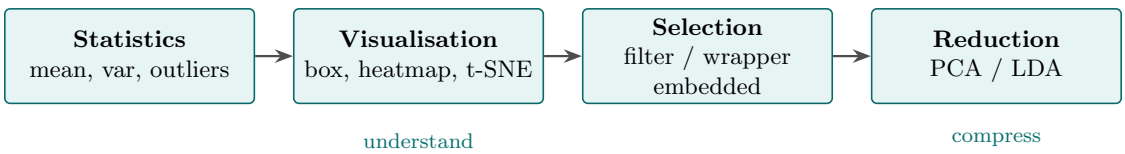


Figure 7: Feature analysis stages.

5.1 Statistical Analysis

- **Mean & variance** — reveal scale differences that require normalisation.
- **Distribution** — checks skew so suitable transforms (e.g. log) can be applied.
- **Outliers** — flags artifact-contaminated windows that survived preprocessing.
- **Correlation** — finds redundant features carrying the same information.

5.2 Visualisation

- **Histogram** — shows the shape of each feature's distribution.
- **Box plot** — compares a feature across the 4 emotion classes.
- **Heatmap** — displays feature–feature correlation at a glance.
- **Scatter plot** — shows how two features separate classes.
- **PCA / t-SNE / UMAP** — projects high-dimensional features to 2D to see class clusters.
- **t-SNE of CNN embeddings** — checks whether the 512-d face features form separate emotion clusters.
- **Grad-CAM** — heatmap of where the CNN looks — should be eyes, brows and mouth, not background.

5.3 Feature Selection (EEG only)

- **Scope** — applied to EEG features only; the CNN selects its own features during training.
- **Correlation filter** — removes one of every pair of highly correlated features.
- **ANOVA F-test** — keeps features whose means differ significantly between classes.
- **Mutual Information** — keeps features that share most information with the label (catches non-linear links).
- **RFE (wrapper)** — repeatedly trains a model and drops the weakest feature.
- **Sequential selection** — adds/removes features one at a time based on validation score.
- **LASSO (embedded)** — L1 penalty pushes useless feature weights to zero.
- **Tree importance** — Random Forest / XGBoost rank features by how much they reduce impurity.

5.4 Dimensionality Reduction

- **PCA** — keeps directions of maximum variance (e.g. 95% variance retained).
- **LDA** — projects onto at most $K - 1 = 3$ axes that best separate the classes.
- **ICA** — finds statistically independent components.
- **Autoencoder** — neural network learns a compact non-linear code.

6 Train / Validation / Test Split

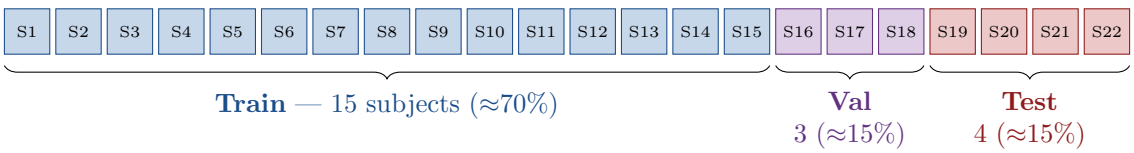


Figure 8: Subject-independent split of the 22 DEAP subjects with face video. A subject never appears in two sets.

- **Subject-independent split** — whole subjects go to one set only, so the model is tested on unseen people.
- **Why not random split** — windows of the same person in train and test leak identity and inflate accuracy.
- **Validation set** — used only for choosing models, MF parameters and Sugeno rules.

- **Test set** — touched exactly once, after everything is frozen.
- **LOSO (optional)** — leave-one-subject-out cross-validation gives a more stable estimate on small data.
- **Fit scalers on train only** — normalisation statistics from val/test would also cause leakage.

7 Individual Models & Choosing the Best Model

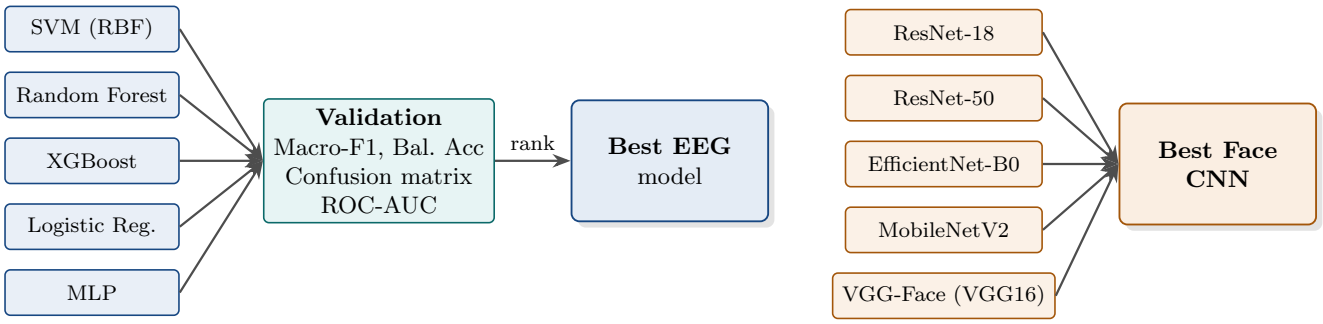


Figure 9: Model selection for each branch (same validation subjects, same metrics).

7.1 EEG Candidate Models (feature-based)

- **SVM (RBF)** — strong baseline for small, high-dimensional EEG feature sets.
- **Random Forest** — robust to noise and gives feature importance for free.
- **XGBoost** — boosted trees; usually the strongest on tabular features.
- **Logistic Regression** — simple, interpretable, naturally calibrated probabilities.
- **MLP** — captures non-linear feature interactions.
- **Why classical ML for EEG** — EEG is already turned into tabular features (PSD, DE); these models suit tabular data with few subjects.

7.2 Face Candidate Models (CNNs)

- **ResNet-18** — light residual network — fast to fine-tune, good default.
- **ResNet-50** — deeper and more accurate, but heavier and easier to overfit on small data.
- **EfficientNet-B0** — best accuracy-per-parameter by scaling depth, width and resolution together.
- **MobileNetV2** — very light; ideal if training on a laptop GPU.
- **VGG-Face (VGG16)** — pretrained on faces, so its features already suit expressions.

7.3 CNN Training Setup

- **Two-phase fine-tuning** — first train only the new head, then unfreeze the last block(s) with a smaller learning rate.
- **Loss: weighted cross-entropy** — class weights stop the model ignoring rare emotion quadrants.
- **Optimiser: AdamW** — Adam with weight decay for stable, regularised training.
- **Cosine LR schedule** — learning rate decays smoothly for better convergence.
- **Early stopping** — stops when validation macro-F1 stops improving, preventing overfitting.
- **Batch norm in eval mode** — keeps pretrained statistics stable when batches are small.

7.4 Evaluation Metrics

- **Accuracy** — fraction of correct predictions — misleading when classes are imbalanced.
- **Precision** — of the samples predicted as an emotion, how many truly are.
- **Recall** — of the true samples of an emotion, how many were found.
- **Macro-F1 (primary)** — harmonic mean of precision and recall, averaged equally over classes.
- **Balanced accuracy** — mean recall per class; fair under imbalance.
- **Confusion matrix** — shows which emotions are mixed up with which.
- **ROC-AUC (one-vs-rest)** — measures ranking quality of the predicted probabilities.

7.5 Selection Rule

- **Select by validation macro-F1** — the model with the best F1 on validation subjects is kept for each branch.
- **Hyper-parameter tuning** — grid / Bayesian search (EEG) and learning rate / unfrozen depth (CNN) on validation data only.
- **Record reliability** ρ_E, ρ_F — the validation F1 of each best model is reused later by the fuzzy rules.

8 Converting Predictions into Crisp Outputs

- **EEG probability output** — use `predict_proba` instead of a hard label.
- **Face probability output** — the CNN's softmax, averaged over frames in the window.
- **Calibration (Platt / isotonic)** — makes EEG "0.8" really mean 80% likely — essential before fuzzy rules.
- **Temperature scaling (CNN)** — divides logits by a learned T to fix the typical over-confidence of CNNs.
- **EEG confidence** c_E — how far the top probability is above random guessing.
- **Face confidence** c_F — same measure for the face model.
- **Agreement** A — overlap of the two probability vectors; 1 = identical, 0 = complete conflict.

$$c_E = \frac{\max_k p_E^k - \frac{1}{K}}{1 - \frac{1}{K}}, \qquad c_F = \frac{\max_k p_F^k - \frac{1}{K}}{1 - \frac{1}{K}}, \qquad A = \sum_{k=1}^{K} \min\left(p_E^k,\, p_F^k\right), \qquad K = 4 \tag{1}$$

- **Why rescale confidence** — with 4 classes the top probability is always ≥ 0.25, so it is mapped to $[0, 1]$.

Model	Happy	Calm	Angry	Sad	Crisp descriptors
EEG $\mathbf{P}_E$	0.72	0.08	0.05	0.15	$c_E = 0.627$
Face $\mathbf{P}_F$	0.85	0.06	0.03	0.06	$c_F = 0.800$
$\min(p_E^k, p_F^k)$	0.72	0.06	0.03	0.06	$A = 0.870$

9 Multimodal Fusion Strategy

- **Early fusion** — concatenate EEG features + CNN embedding into one model — simple but sensitive to scale and dimension.

- **Late fusion (chosen)** — combine model outputs — modular, interpretable, and a natural fit for fuzzy rules.
- **Hybrid fusion** — mixes intermediate features and outputs — most powerful but most complex.
- **Baselines** — early fusion and simple probability averaging are kept for comparison with Sugeno.

10 Sugeno Fuzzy Inference System

- **Why Sugeno** — consequents are numbers/functions, so output is crisp and computationally cheap — no centroid needed.
- **Why two stages** — FIS-1 decides *whom to trust*; FIS-2 decides *which emotion*, using that trust.
- **Why not one numeric emotion code** — averaging Happy = 1 and Angry = 3 gives Sad = 2 — meaningless, so each emotion gets its own score.

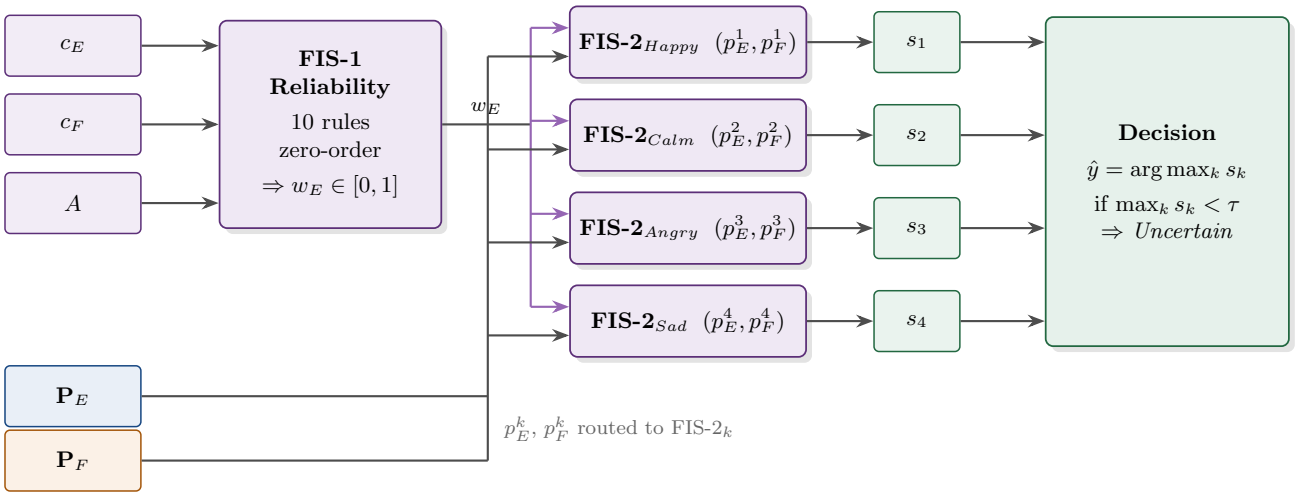


Figure 10: Two-stage Sugeno fusion. FIS-1 outputs a trust weight w_E (face trust $= 1 - w_E$); four parallel FIS-2 blocks score each emotion; the highest score is the prediction.

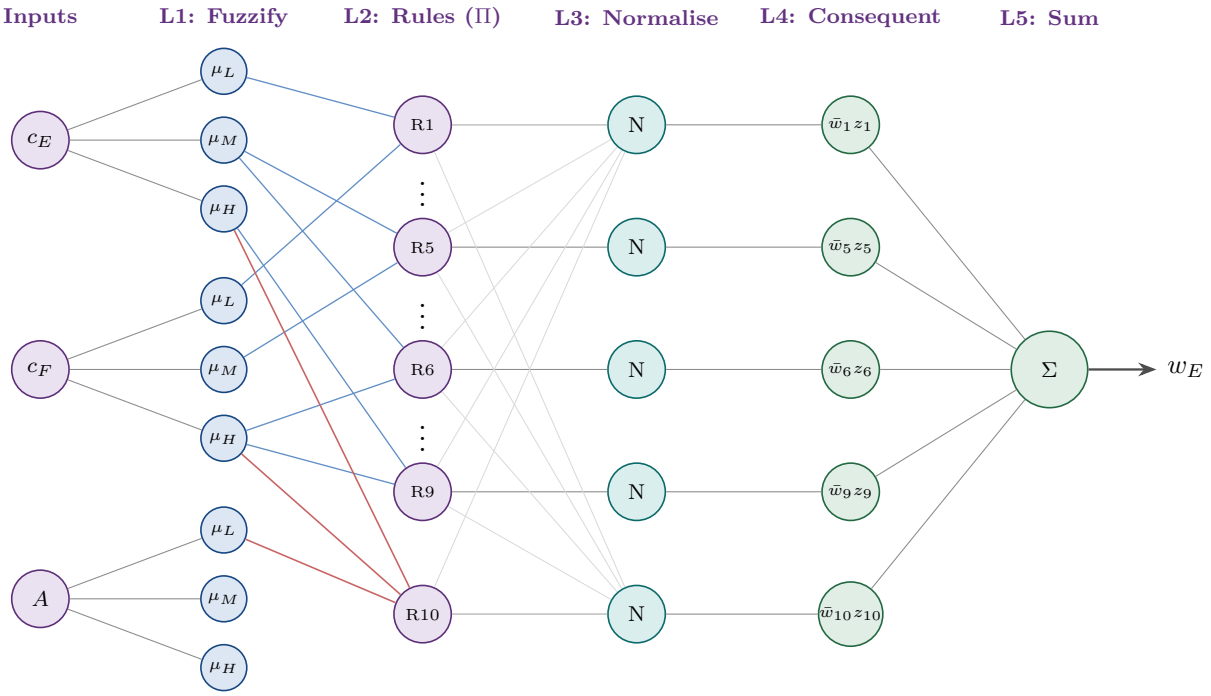


Figure 11: FIS-1 drawn as an adaptive (ANFIS-style) network. L1 fuzzifies inputs, L2 computes rule firing strengths by product, L3 normalises them, L4 multiplies by rule constants, L5 sums. The red rule R10 uses agreement A to resolve conflicts.

10.1 Fuzzy Inputs and Outputs

System	Inputs	Linguistic terms	Output
FIS-1	c_E, c_F, A	Low, Medium, High	w_E = trust in EEG (face trust = $1 - w_E$)
FIS-2_k	p_E^k, p_F^k (+ w_E as parameter)	Low, Medium, High	s_k = score of emotion k

- **Few inputs** — only 2–3 inputs per FIS keep the rule base small ($3 \times 3 = 9$ core rules).
- w_E **as a parameter** — FIS-1's output feeds FIS-2 constants, linking the two stages.

10.2 Membership Functions

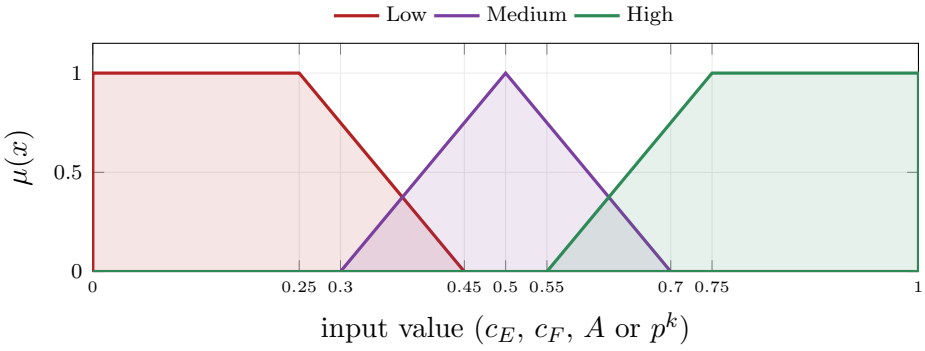


Figure 12: Initial membership functions (shared by all inputs; tuned later on validation data).

$$\mu_L(x) = \text{trap}(x; 0, 0, 0.25, 0.45), \quad \mu_M(x) = \text{tri}(x; 0.30, 0.50, 0.70), \quad \mu_H(x) = \text{trap}(x; 0.55, 0.75, 1, 1) \tag{2}$$

- **Trapezoid at the edges** — values near 0 or 1 are fully Low or fully High.
- **Triangle in the middle** — Medium peaks at 0.5 and fades on both sides.

- **Overlap** — neighbouring terms overlap so a value can be partly Medium and partly High — the essence of fuzziness.
- **Initial values only** — the break-points are optimised on validation data (Sec. 11).

10.3 Rule Base A — FIS-1 (Modality Reliability)

- **Goal** — output a trust weight w_E: high when EEG is the more confident/reliable modality.
- **Consequents** — zero-order constants; $w_E = 0.5$ means equal trust.

	Face confidence c_F		
EEG conf. c_E	**Low**	**Medium**	**High**
Low	R1: 0.50	R2: 0.30	R3: 0.15
Medium	R4: 0.70	R5: 0.50	R6: 0.35
High	R7: 0.85	R8: 0.65	R9: 0.50

■ trust EEG ■ trust Face ■ balanced

FIS-1 rules (output w_E = trust in EEG)

R1	IF c_E is **Low** AND c_F is **Low** THEN $w_E = 0.50$	*both weak → no preference*
R2	IF c_E is **Low** AND c_F is **Medium** THEN $w_E = 0.30$	*lean on face*
R3	IF c_E is **Low** AND c_F is **High** THEN $w_E = 0.15$	*trust face*
R4	IF c_E is **Medium** AND c_F is **Low** THEN $w_E = 0.70$	*lean on EEG*
R5	IF c_E is **Medium** AND c_F is **Medium** THEN $w_E = 0.50$	*balanced fusion*
R6	IF c_E is **Medium** AND c_F is **High** THEN $w_E = 0.35$	*lean on face*
R7	IF c_E is **High** AND c_F is **Low** THEN $w_E = 0.85$	*trust EEG*
R8	IF c_E is **High** AND c_F is **Medium** THEN $w_E = 0.65$	*lean on EEG*
R9	IF c_E is **High** AND c_F is **High** THEN $w_E = 0.50$	*strong agreement*
R10	IF c_E is **High** AND c_F is **High** AND A is **Low** THEN $w_E = \omega^* = \dfrac{\rho_E}{\rho_E + \rho_F}$	*confident conflict → trust the historically better model*

- **R1–R9** — trust follows relative confidence: the more confident modality gets more weight.
- **R10 (conflict rule)** — when both are confident but disagree, fall back on validated reliability ρ (validation F1).

FIS-1 inference.

$$w_i = \prod_j \mu_{ij}(x_j) \quad \text{(product AND)}, \qquad w_E = \frac{\sum_{i=1}^{10} w_i\, z_i}{\sum_{i=1}^{10} w_i} \tag{3}$$

- **Firing strength** w_i — how strongly the inputs satisfy rule i.
- **Weighted average** — the Sugeno defuzzifier: rule constants z_i averaged by firing strength.

10.4 Rule Base B — FIS-2 (Emotion-Specific Scoring)

- **Goal** — for each emotion k, produce a support score $s_k \in [0, 1]$ from both modalities.
- **Same 9 rules for every emotion** — only the inputs p_E^k, p_F^k change, keeping the system compact.

	Face probability p_F^k		
EEG prob. p_E^k	**Low**	**Medium**	**High**
Low	B1: 0	B2: $0.6(1 - w_E)$	B3: $1 - w_E$
Medium	B4: $0.6\, w_E$	B5: 0.55	B6: 0.80
High	B7: w_E	B8: 0.80	B9: 1.00

FIS-2$_k$ rules (output s_k = support for emotion k)

B1	IF p_E^k is **Low** AND p_F^k is **Low** THEN $s_k = 0$	*neither supports k*
B2	IF p_E^k is **Low** AND p_F^k is **Medium** THEN $s_k = 0.6(1 - w_E)$	*weak face-only support*
B3	IF p_E^k is **Low** AND p_F^k is **High** THEN $s_k = 1 - w_E$	*face-only, scaled by face trust*
B4	IF p_E^k is **Medium** AND p_F^k is **Low** THEN $s_k = 0.6\, w_E$	*weak EEG-only support*
B5	IF p_E^k is **Medium** AND p_F^k is **Medium** THEN $s_k = 0.55$	*moderate joint support*
B6	IF p_E^k is **Medium** AND p_F^k is **High** THEN $s_k = 0.80$	*strong joint support*
B7	IF p_E^k is **High** AND p_F^k is **Low** THEN $s_k = w_E$	*EEG-only, scaled by EEG trust*
B8	IF p_E^k is **High** AND p_F^k is **Medium** THEN $s_k = 0.80$	*strong joint support*
B9	IF p_E^k is **High** AND p_F^k is **High** THEN $s_k = 1.00$	*full agreement*

- **Agreement rules (B5, B6, B8, B9)** — both modalities support k → high fixed score.
- **Single-modality rules (B2, B3, B4, B7)** — only one modality supports k → score scaled by that modality's trust.
- **B1** — no support from either modality → zero score.

10.5 Emotion-Specific Rules (Expanded)

- **Expansion** — Rule Base B instantiated for each DEAP class gives the readable emotion rules below.

Happy (HV–HA)

H1 IF EEG predicts Happy is **High** AND Face predicts Happy is **High** THEN $s_{\text{Happy}} = 1.00$
H2 IF EEG predicts Happy is **Medium** AND Face predicts Happy is **High** THEN $s_{\text{Happy}} = 0.80$
H3 IF EEG predicts Happy is **Low** AND Face predicts Happy is **High** THEN $s_{\text{Happy}} = 1 - w_E$

Calm (HV–LA)

C1 IF EEG predicts Calm is **High** AND Face predicts Calm is **High** THEN $s_{\text{Calm}} = 1.00$
C2 IF EEG predicts Calm is **High** AND Face predicts Calm is **Low** THEN $s_{\text{Calm}} = w_E$ *calm faces are subtle; EEG alpha often decides*
C3 IF EEG predicts Calm is **Medium** AND Face predicts Calm is **Medium** THEN $s_{\text{Calm}} = 0.55$

Angry (LV–HA)

A1 IF EEG predicts Angry is **High** AND Face predicts Angry is **High** THEN $s_{\text{Angry}} = 1.00$
A2 IF EEG predicts Angry is **High** AND Face predicts Angry is **Medium** THEN $s_{\text{Angry}} = 0.80$
A3 IF EEG predicts Angry is **Low** AND Face predicts Angry is **Medium** THEN $s_{\text{Angry}} = 0.6(1 - w_E)$

Sad (LV–LA)

S1 IF EEG predicts Sad is **High** AND Face predicts Sad is **High** THEN $s_{\text{Sad}} = 1.00$
S2 IF EEG predicts Sad is **Medium** AND Face predicts Sad is **High** THEN $s_{\text{Sad}} = 0.80$
S3 IF EEG predicts Sad is **High** AND Face predicts Sad is **Low** THEN $s_{\text{Sad}} = w_E$

Conflict and Uncertainty

D1 IF EEG predicts Happy is **High** AND Face predicts Sad is **High** THEN $s_{\text{Happy}} = w_E$ and $s_{\text{Sad}} = 1 - w_E$ *(B7 + B3) — more trusted modality wins*
D2 IF c_E is **High** AND c_F is **High** AND A is **Low** THEN $w_E = \omega^*$ *(R10) — validated reliability decides*
U1 IF c_E is **Low** AND c_F is **Low** THEN all s_k stay small ⇒ $\max_k s_k < \tau$ ⇒ **Uncertain**

- **Full rule count** — 9 rules × 4 emotions = 36 FIS-2 rules + 10 FIS-1 rules = 46 rules.

10.6 Final Decision

$$s_k = \frac{\sum_{i=1}^{9} w_{ik}\, z_{ik}}{\sum_{i=1}^{9} w_{ik}}, \qquad \hat{y} = \begin{cases} \arg\max_k s_k, & \max_k s_k \geq \tau \\ \text{Uncertain}, & \text{otherwise} \end{cases} \tag{4}$$

- **Per-emotion score** s_k — weighted average of the 9 FIS-2 rules for emotion k.
- **Arg-max** — the emotion with the strongest fuzzy support is the prediction.
- **Threshold** τ — rejects weak predictions instead of guessing; tuned on validation (e.g. $\tau = 0.5$).
- **Confidence output** — $\max_k s_k$ is reported alongside the emotion.

10.7 Worked Examples

	Example 1: Agreement	**Example 2: Conflict**
$\mathbf{P}_E$ **(H,C,A,S)**	[0.72, 0.08, 0.05, 0.15]	[0.70, 0.05, 0.05, 0.20]
$\mathbf{P}_F$ **(H,C,A,S)**	[0.85, 0.06, 0.03, 0.06]	[0.08, 0.04, 0.08, 0.80]
c_E, c_F, A	0.627, 0.800, 0.870	0.600, 0.733, 0.370
Fuzzified	c_E: M 0.37, H 0.38; c_F: H 1; A: H 1	c_E: M 0.50, H 0.25; c_F: H 0.92; A: L 0.40, M 0.35
Rules fired (FIS-1)	R6 (0.37), R9 (0.38)	R6 (0.46), R9 (0.23), R10 (0.09)
w_E	$\frac{0.37(0.35)+0.38(0.50)}{0.75} = \mathbf{0.43}$	$\frac{0.46(0.35)+0.23(0.50)+0.09(0.45)}{0.78} = \mathbf{0.41}$ (ω^*=0.45 assumed)
FIS-2 scores	$s_H = 1.00$ (B9), others $= 0$	$s_H = w_E = 0.41$ (B7), $s_S = 1 - w_E = 0.59$ (B3)
Output	**Happy**, confidence 1.00	**Sad**, confidence 0.59

- **Example 1** — both agree on Happy → rule B9 fires fully → confident Happy.
- **Example 2** — EEG says Happy, face says Sad; face is more confident → Sad wins with moderate confidence.

11 Optimising the Sugeno System

Rule set	**Strategy**	**FIS-1 constants**
A	Equal weighting	all $z_i = 0.5$
B	EEG-weighted	all z_i shifted $+0.15$
C	Face-weighted	all z_i shifted -0.15
D	Confidence-dependent	table in Rule Base A (proposed)

- **Compare rule sets A–D** — pick the one with best validation macro-F1 and balanced accuracy.
- **Tune MF break-points** — grid search over Low/Medium/High boundaries.
- **Tune constants** z_i **and** τ — small search (or gradient/ANFIS learning) on validation data.
- **Never tune on test data** — test performance must stay an honest estimate.

12 Final Testing & Reporting

- **Freeze** — preprocessing, features, models, MFs, rules and τ are all fixed.
- **Single test run** — evaluate once on the 4 unseen test subjects.
- **Ablation** — compare each step so the gain from Sugeno is clearly shown.

System	**Accuracy**	**Macro-F1**	**Bal. Acc.**	**Coverage**
EEG only	–	–	–	100%
Face baseline (landmarks/AU + SVM)	–	–	–	100%
Face only (fine-tuned CNN)	–	–	–	100%
Early fusion (concat)	–	–	–	100%
Late fusion (prob. average)	–	–	–	100%
EEG + Face + Sugeno (proposed)	–	–	–	–

Fill after the final test run. Coverage = % of samples not labelled "Uncertain".

- **Coverage** — reported because the Sugeno system may reject low-confidence samples.
- **Confusion matrices** — shown for EEG-only, Face-only and Sugeno to explain where fusion helps.

13 Conclusion

EEG → preprocessing → features → best classifier → $\mathbf{P}_E$
Face → preprocessing → fine-tuned CNN → softmax → window average → $\mathbf{P}_F$
$(\mathbf{P}_E, \mathbf{P}_F)$ → **FIS-1** (whom to trust) → **FIS-2** (which emotion) → **final emotion + confidence**

- **Interpretable** — every decision can be traced to named IF–THEN rules.
- **Robust** — a noisy or uncertain modality is automatically down-weighted.
- **Honest** — subject-independent evaluation and an "Uncertain" option avoid inflated results.